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A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere with North and South America. The image is framed by a thin white border.

COMP-2024 / COMP-2039
**Artificial Intelligence
Methods**

Lecture 6b
**Benchmark Function
Optimisation**



Learning Outcomes

IDENTIFY

1. Travelling Salesman Problem
2. Generic Permutation based Genetic Operators
3. Multimeme Memetic Algorithms
4. **Benchmark Function Optimisation**



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Benchmark Function Optimisation



Benchmark Function Optimisation

Label	Function name	lb	ub	opt	<i>isContinuous</i>	<i>isSeparable</i>	<i>isMultimodal</i>
F1	Sphere	-5.12	5.12	0	yes	yes	no
F2	Rosenbrock	-2.048	2.048	0	yes	yes	no
F3	Step	-5.12	5.12	0	yes	yes	no
F4	Quartic <i>with noise</i>	-1.28	1.28	1	yes	yes	yes
F5	Foxhole	-65.536	65.536	1	yes	no	yes
F6	Rastrigin	-5.12	5.12	0	yes	yes	yes
F7	Schwefel	-500	500	0	yes	yes	yes
F8	Griewangk	-600	600	0	yes	no	yes
F9	Ackley	-32.768	32.768	0	yes	no	yes
F10	Easom	-100	100	-1	yes	no	no
F11	Schwefel's Double Sum	-65.536	65.536	0	yes	no	no
F12	Royal Road	-	-	0	no	yes	n/a
F13	Goldberg	-	-	0	no	yes	n/a
F14	Whitley	-	-	0	no	yes	n/a

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<http://www.cs.nott.ac.uk/~pszeo/docs/publications/IDA08.pdf>



A 3D surface plot of the Exton function. The plot shows a dark, saddle-shaped surface with a prominent sharp peak (colored red and yellow) and a sharp valley (colored blue) near the center. The axes are labeled x_1 , x_2 , and $f(x_1, x_2)$. The x_1 and x_2 axes range from -20 to 20, and the $f(x_1, x_2)$ axis ranges from -1 to 0.4. The title of the plot is "Exton function".

5



Binary vs Gray Encoding

- *Gray encoding* ensures a Hamming distance of 1 for the adjacent numbers.
- Shown to be useful in GAs empowering the algorithm to mutate a solution in the right direction.

Dec	Binary	Gray
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100



Components of Evolutionary Algorithms and Settings

- **Representation:** Gray encoding
- **Initialisation:** random
- **Mate selection:** Tournament with tour size of 2.
- **Crossover:** 1PTX ($p_c=1.0$)
- Traditional mutation based on bit-flip with mutation rate $1/(2 \times \text{chromosome_length})$.
- A trans-generational EA with a replacement method which keeps only two best individuals from the previous generation.

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<i>Label</i>	<i>dim</i>	<i>bits</i>	<i>Chrom. Len.</i>	<i>Pop. size</i>	<i>Max. hcteps</i>
F1	10	30	300	60	600
F2	10	30	300	60	600
F3	10	30	300	60	600
F4	10	30	300	60	600
F5	2	30	60	20	120
F6	10	30	300	60	600
F7	10	30	300	60	600
F8	10	30	300	60	600
F9	10	30	300	60	600
F10	6	30	180	36	360
F11	10	30	300	60	600
F12	8	8	64	20	128
F13	30	3	90	20	180
F14	6	4	24	20	48



Components of Evolutionary Algorithms and Settings

- Memes (using bit-flip):
 - GA: Genetic Algorithm (none)
 - MA0: MA with SDHC (best imp.)
 - MA1: MA with NDHC (first imp.)
 - MA2: MA with RMHC
 - MA3: MA with DBHC
- Expectedly, poor performing memes (using biased moves) are designed:
 - MA4: SDHC $\Rightarrow$ neighborhood move: AND with 0
 - MA5: SDHC $\Rightarrow$ neighborhood move: OR with 1
 - MA6: NDHC $\Rightarrow$ neighborhood move: AND with 0
 - MA7: NDHC $\Rightarrow$ neighborhood move: OR with 1

<i>Label</i>	<i>dim</i>	<i>bits</i>	<i>Chrom. Len.</i>	<i>Pop. size</i>	<i>Max. hcteps</i>
F1	10	30	300	60	600
F2	10	30	300	60	600
F3	10	30	300	60	600
F4	10	30	300	60	600
F5	2	30	60	20	120
F6	10	30	300	60	600
F7	10	30	300	60	600
F8	10	30	300	60	600
F9	10	30	300	60	600
F10	6	30	180	36	360
F11	10	30	300	60	600
F12	8	8	64	20	128
F13	30	3	90	20	180
F14	6	4	24	20	48

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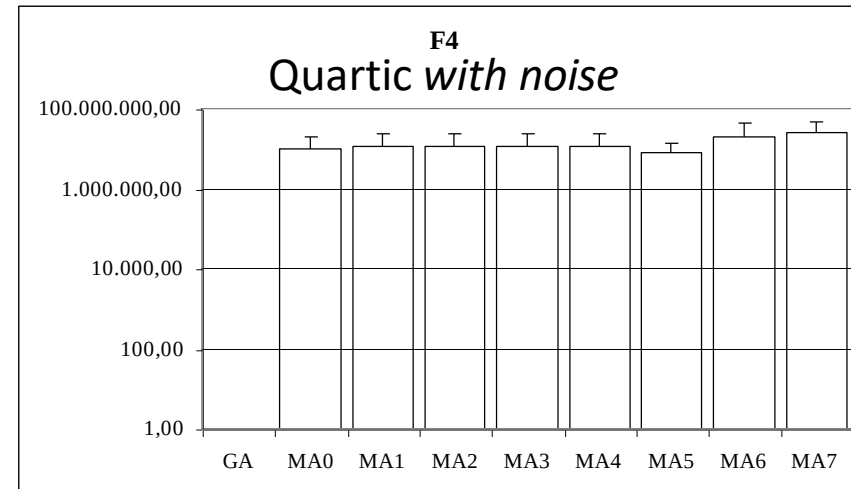
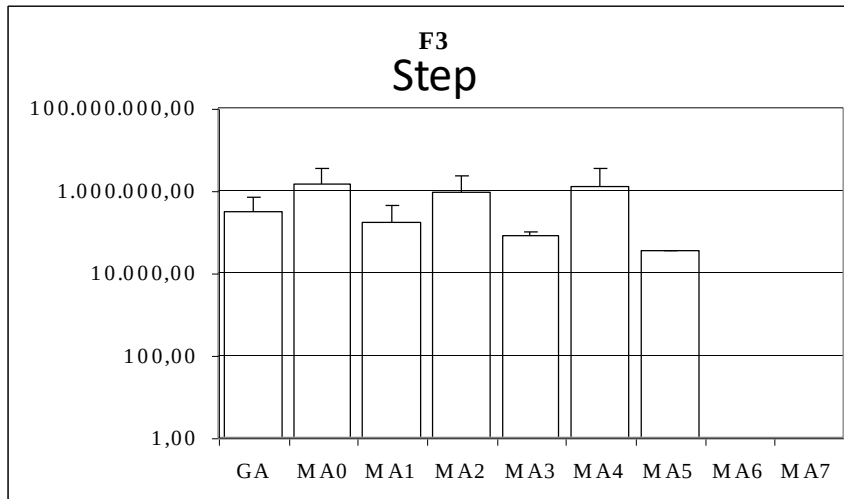
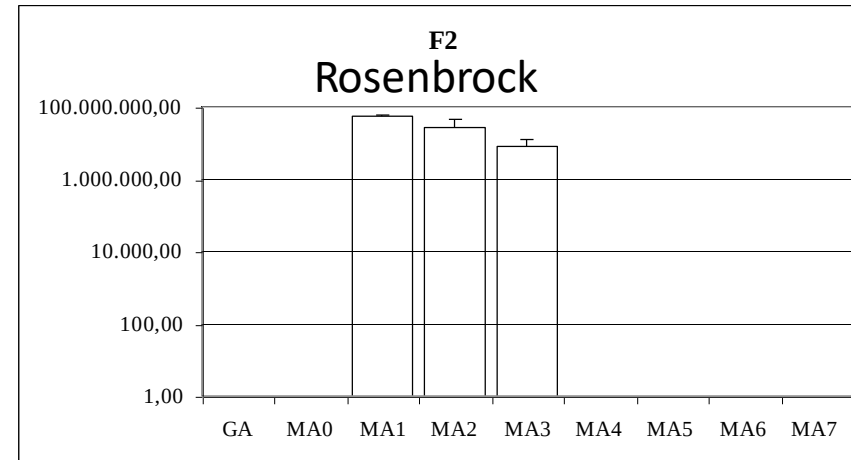
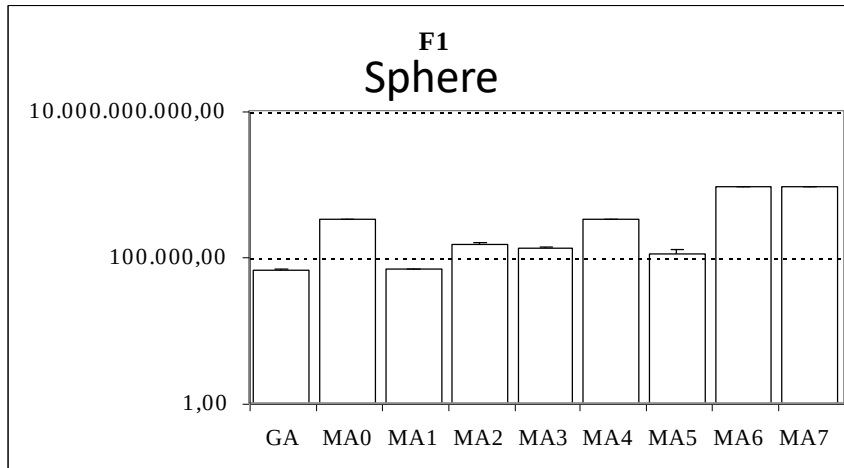


Termination and Performance Comparison Criteria

- **Termination:** Runs are terminated whenever the overall CPU time exceeds 600 sec., or an expected fitness (optimum) is achieved.
 - Pentium IV 2 GHz. machines with 256 MB RAM are used.
- All runs are repeated 50 times.
- Performance indicators:
 - Success rate (effectiveness): The ratio of the number of runs returning the expected optimal solution to the total number of runs (50).
 - Average number of evaluations/configurations (efficiency).
 - Bar chart plots showing the average no. of evaluations in log scale for each algorithm, if the success rate is 100% (all 50 runs yield expected optimum), otherwise no bar is drawn.



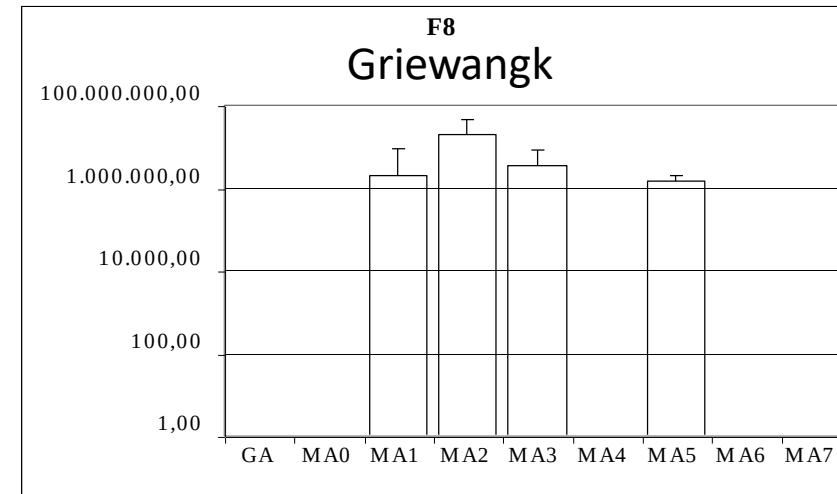
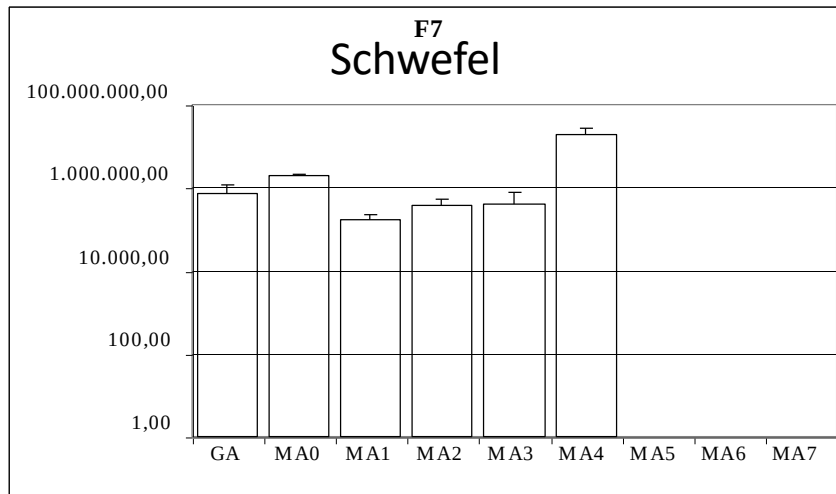
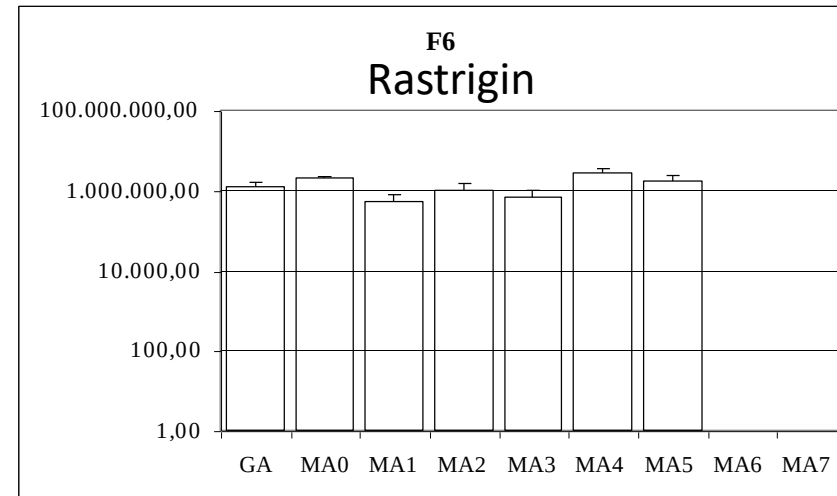
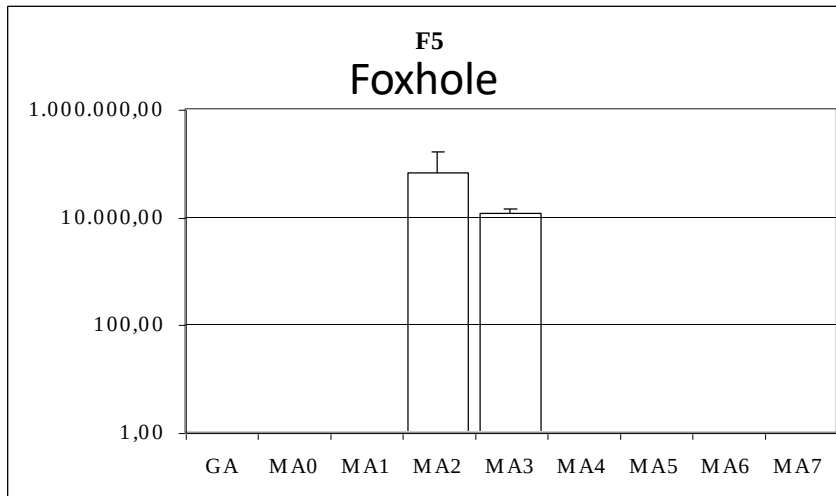
Results



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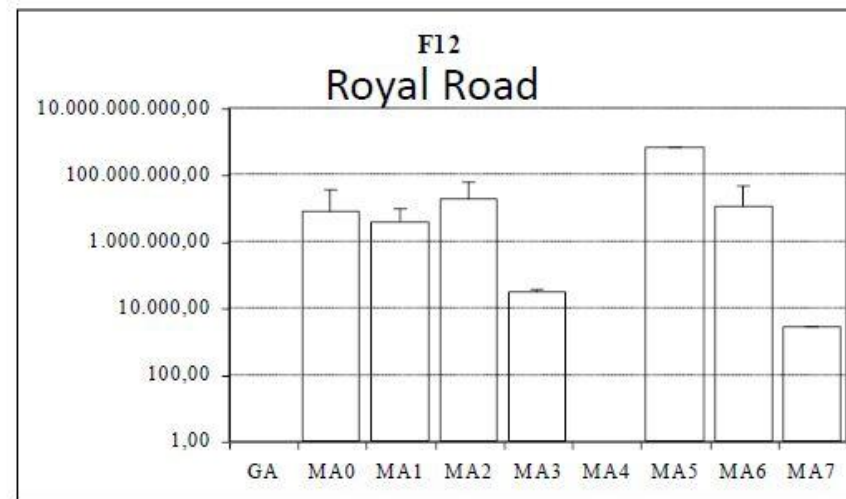
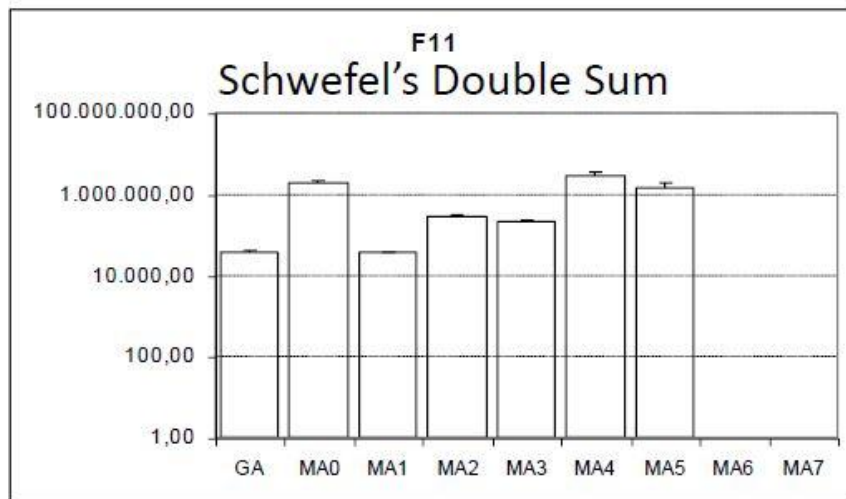
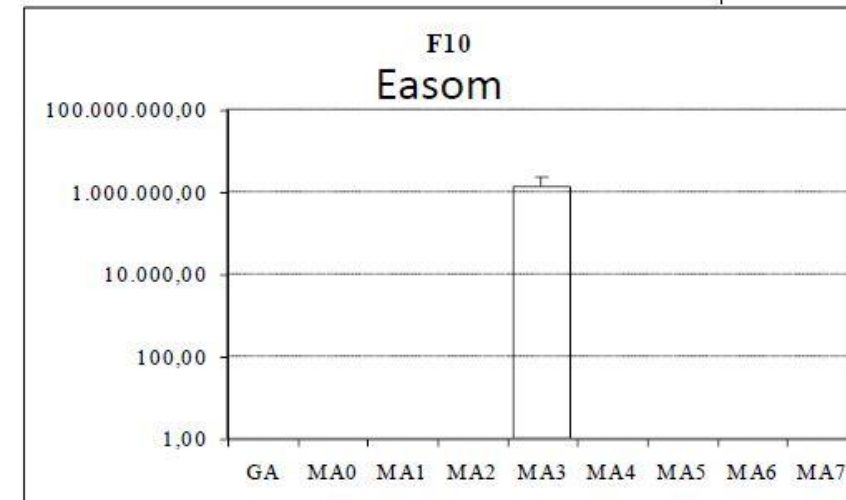
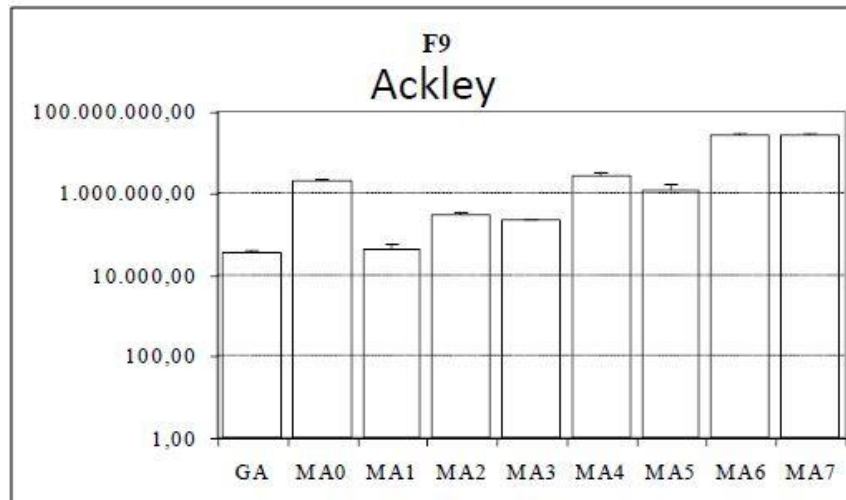
Results



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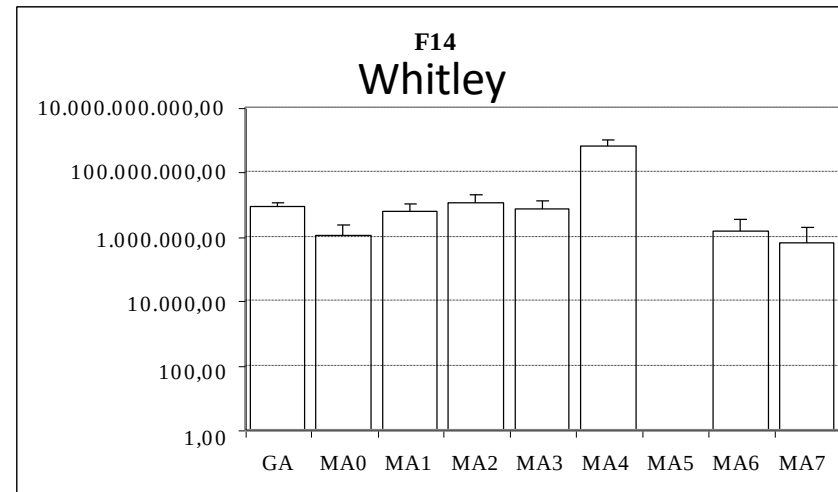
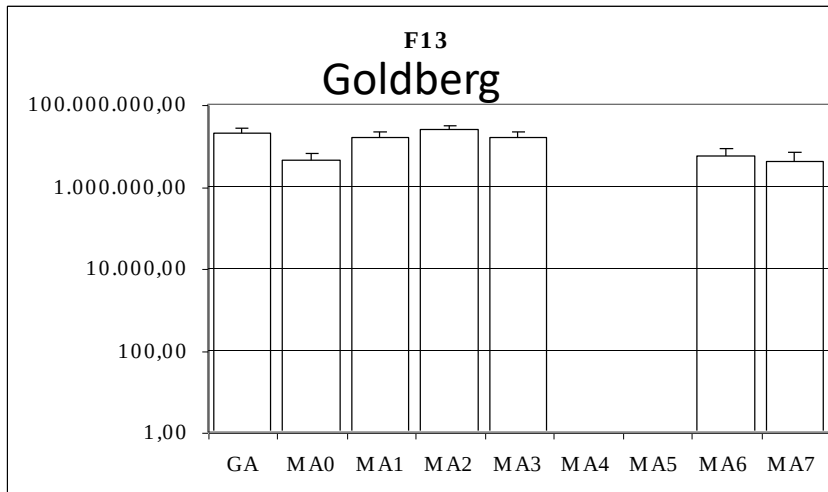
Results



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Results



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Overall Summary

- MA0 (SDHC) is the best meme choice for F4, F13 and F14. (noisy + deceptive)
- MA1 (NDHC) is the best meme choice for F6-F8. (multimodal)
- MA3 is (DBHC) the best meme choice for F2, F3, F5, F10, F12. (functions with plateaus)
- For functions F1, F11 (unimodal) and F9, GA performs slightly better than the memetic algorithm with the meme MA1 (NDHC). The performance difference is insignificant.
- In the overall, MA2 (RMHC) and MA3 (DBHC) turn out to be the worst and the best meme, respectively, among MA0-MA3, considering the success rates and average number of evaluations.
- Different memes yield different performances, designing the right meme for the problem in hand is important.

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Benchmark Function Optimisation - Revisited



MMA Experiments for Benchmark Function Optimisation – Additional Settings

- IR rate is fixed as 0.20.
- Two sets of experiments are performed.

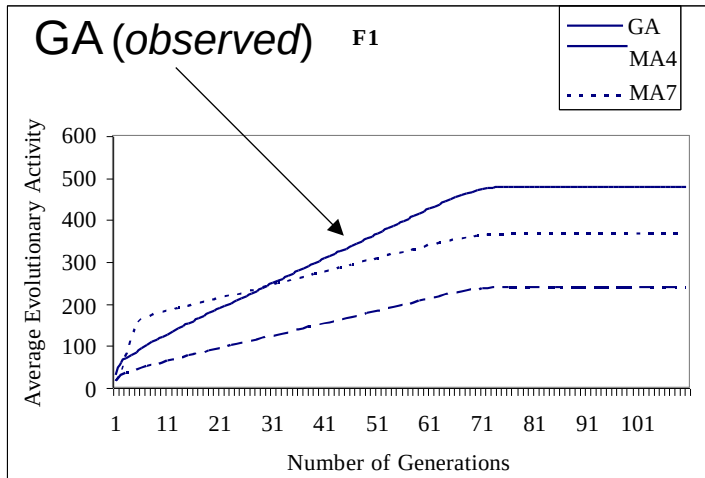
MMA 1. A single *good* meme and two *poor performing* memes are used to test the power of MMAs in identifying the *good* ones.

MMA 2. Memes GA, MA0, MA1, MA2 and MA3 are used to observe whether MMA will provide some type of synergy between memes.

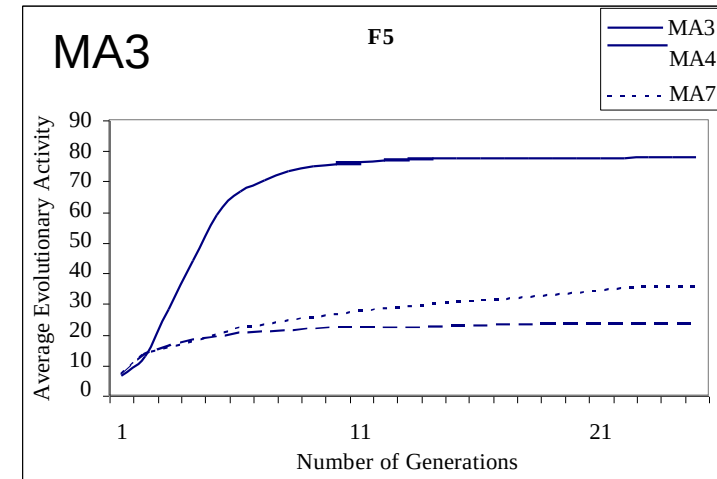
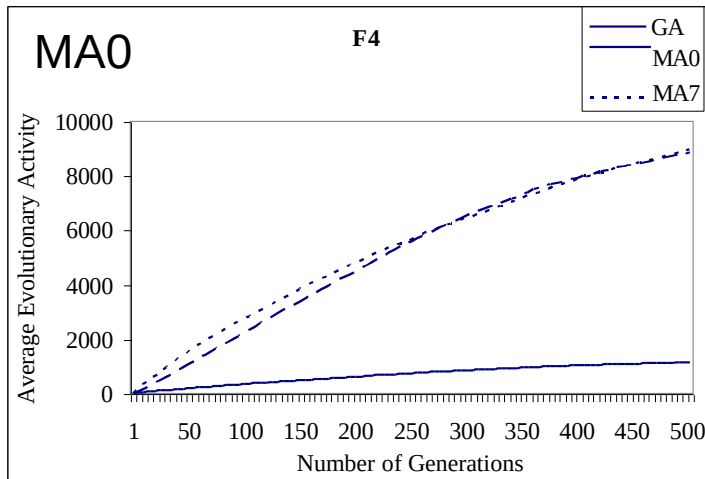
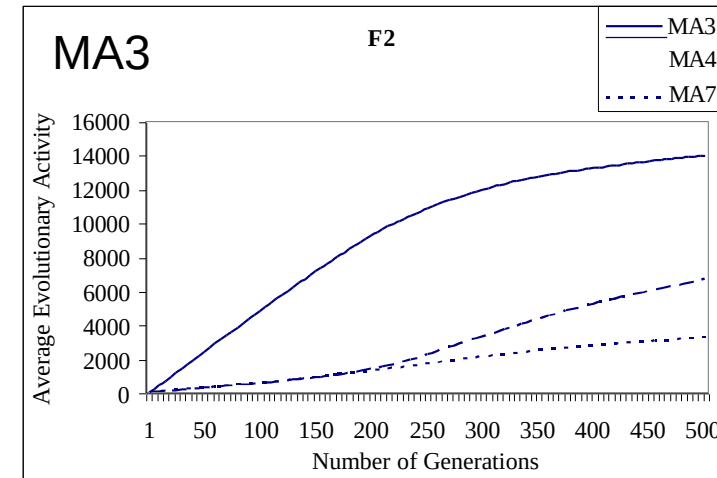


MMA 1.a

GA (expected – good meme)



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MA0

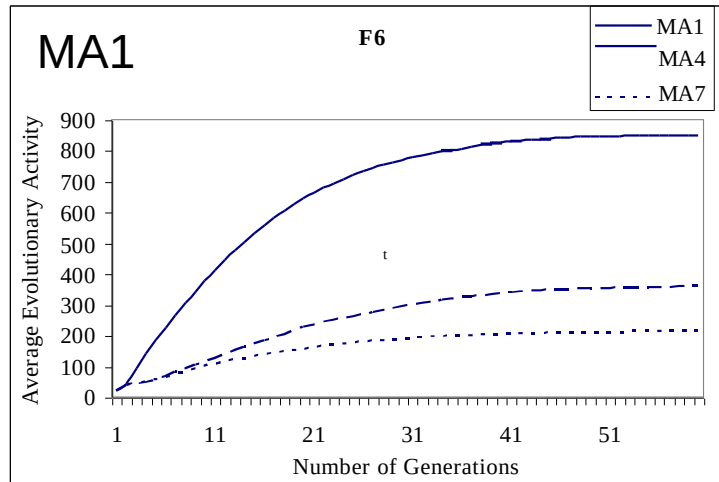
MA3

45

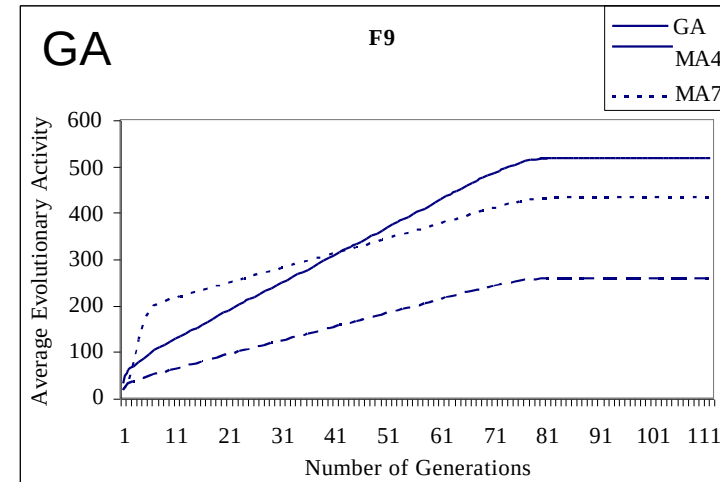
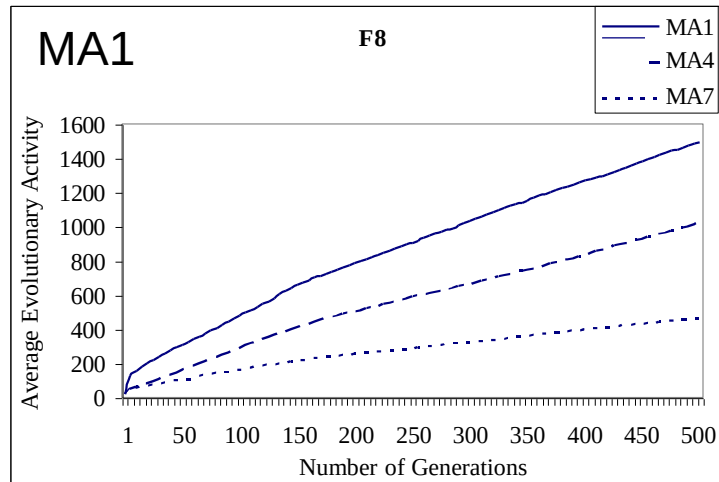
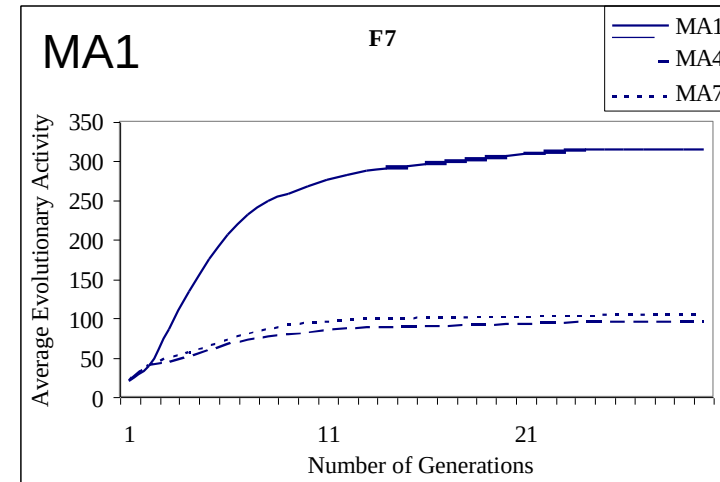


MMA 1.b

MA1



MA1



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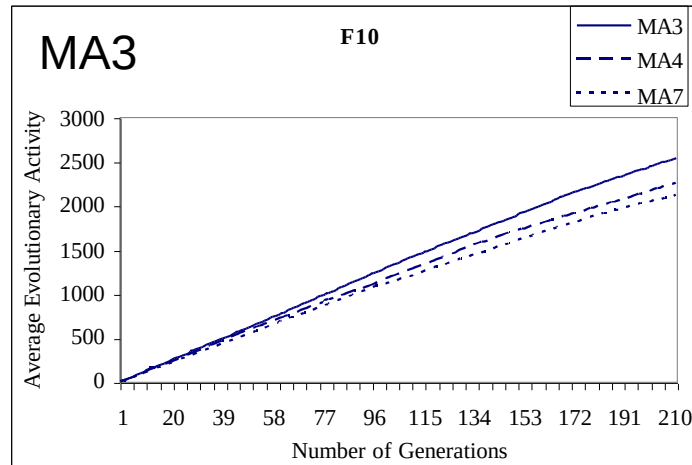
MA1

GA

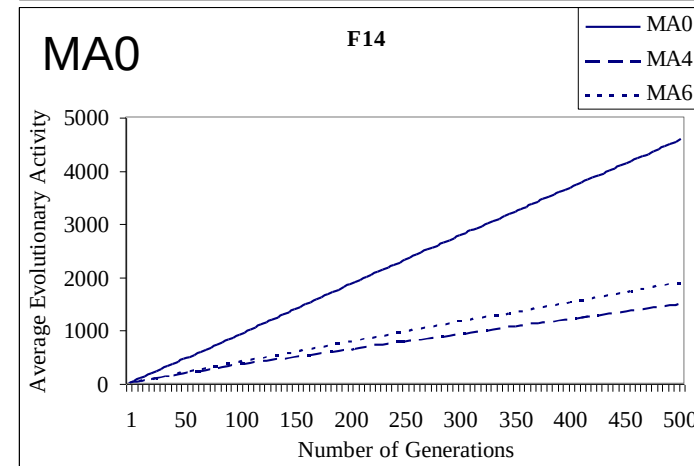
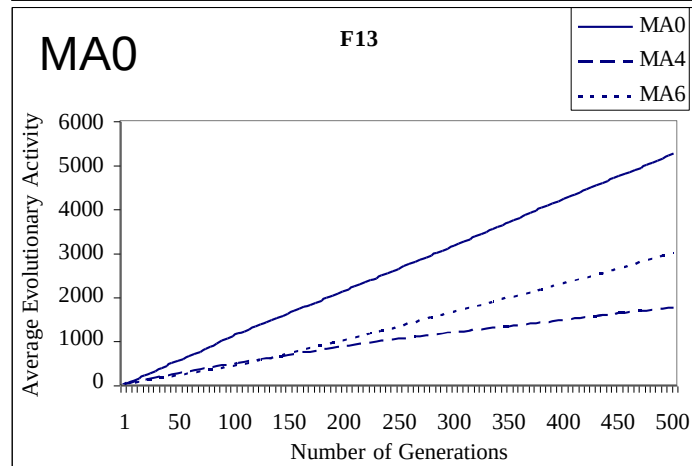
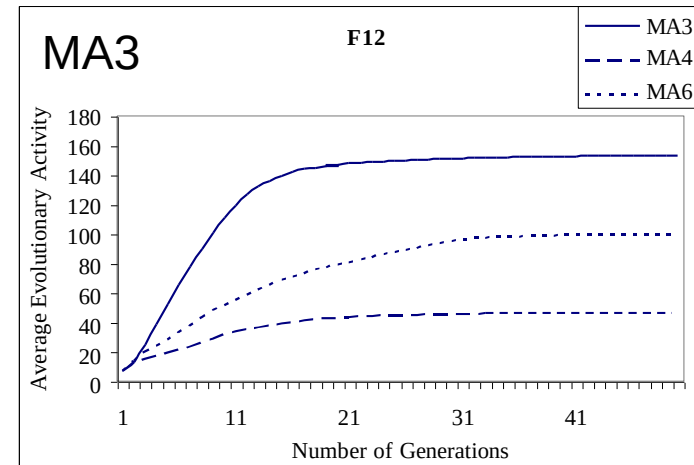


MMA 1.c

MA3



MA3



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MA0

MA0



MMA 2 Experiments – Synergy Between Memes

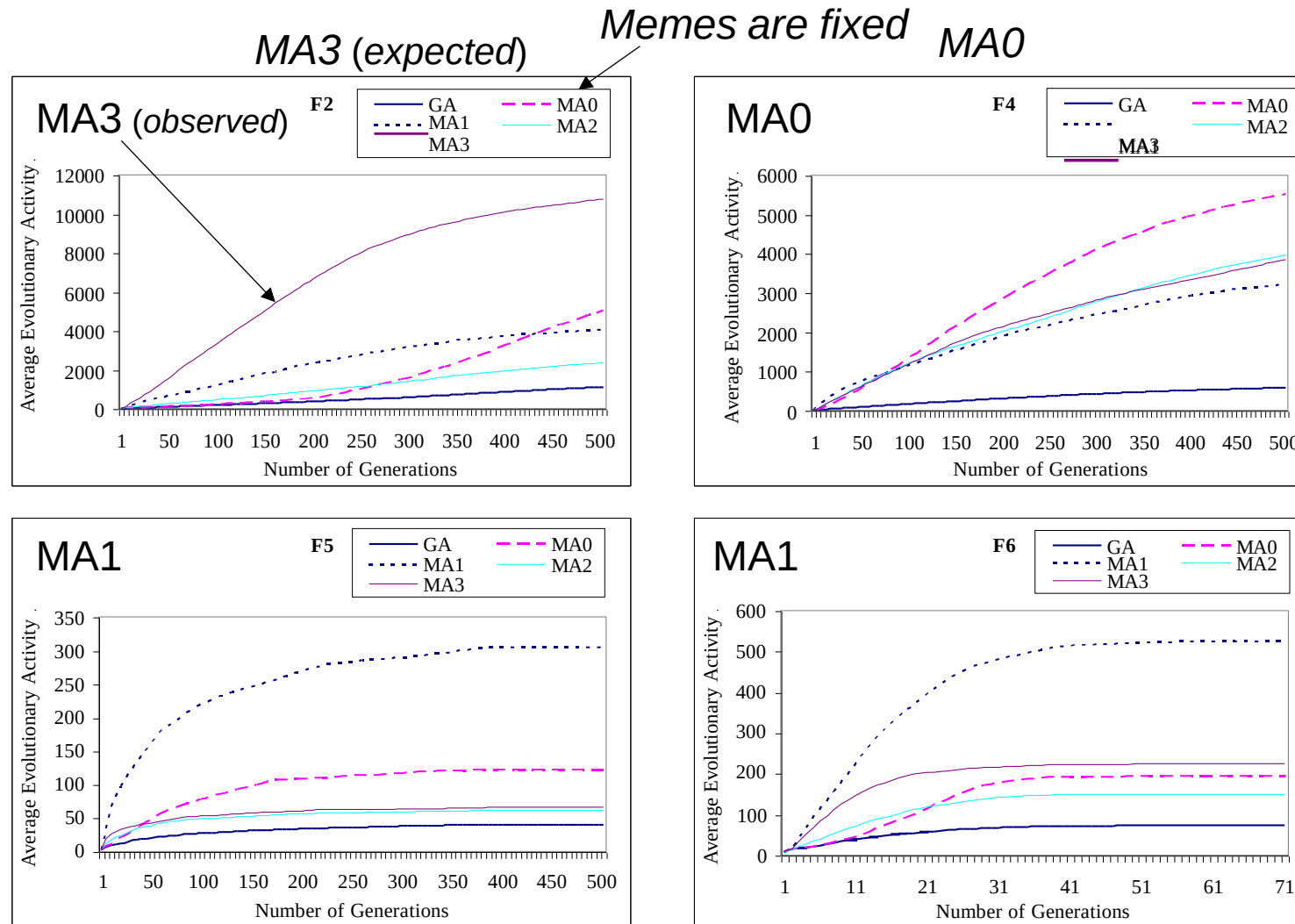
<i>label</i>	<i>type</i>	<i>avr. no. of evals.</i>	<i>st.dev.</i>	<i>label</i>	<i>type</i>	<i>avr. no. of evals.</i>	<i>st.dev.</i>
F1	MMA	17,580	2,226	F8	MMA	5,215,787	9,658,230
	MA1	92,256	0		MA1	1,906,134	6,646,991
F2	MMA	23,605,004	24,364,979	F9	MMA	43,871	12,193
	MA3	8,455,507	3,803,504		MA1	180,783	12,647
F3	MMA	72,252	11,772	F10	MMA	3,100,515	4,565,736
	MA3	82,769	16,512		MA3	1,340,811	988,971
F4	MMA	12,926,879	11,435,876	F11	MMA	17,580	2,226
	MA0	9,494,844	10,332,574		MA1	36,060	0
F5	MMA	46,975	79,394	F12	MMA	31,297	14,961
	MA3	11,619	2,293		MA3	29,246	4,936
F6	MMA	553,306	231,124	F13	MMA	7,667,352	2,832,376
	MA1	525,398	262,055		MA0	4,348,896	1,617,951
F7	MMA	349,250	324,544	F14	MMA	3,674,932	2,623,300
	MA1	167,799	60,577		MA0	1,072,117	1,111,825

Second entry for each function indicates the best performing hill climbing under the generic MA framework where HC is applied after mutation.

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MMA 2.a



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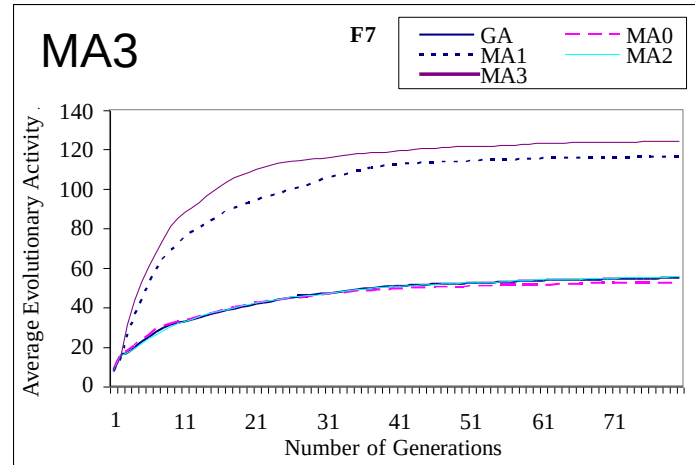
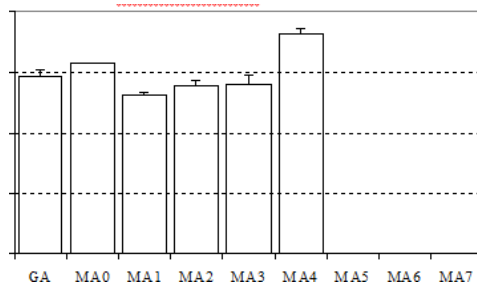


MMA 2.b

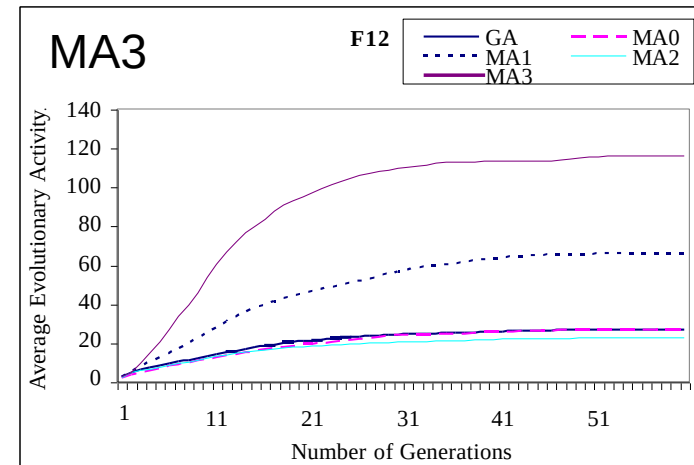
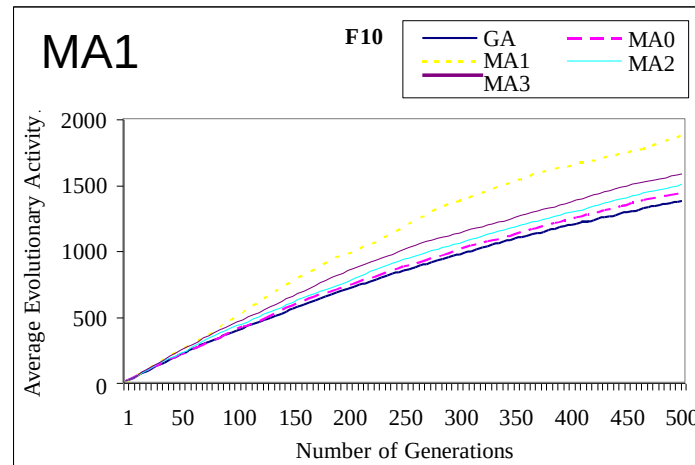
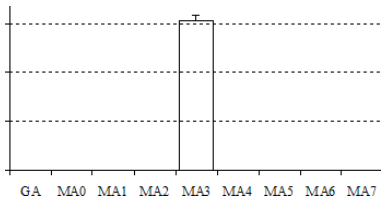
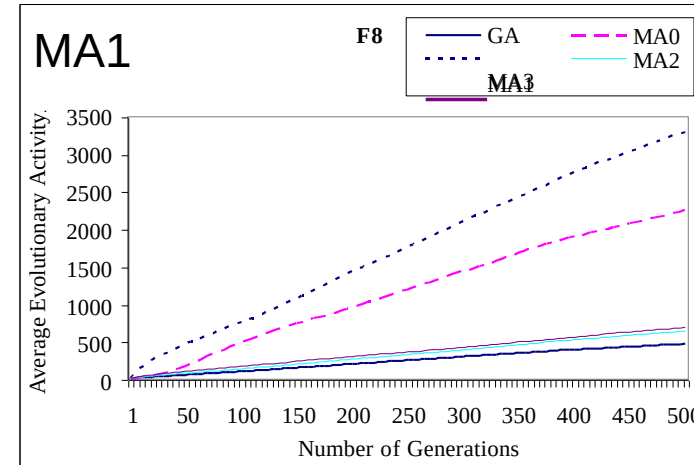
F7

F10

MA1



MA1



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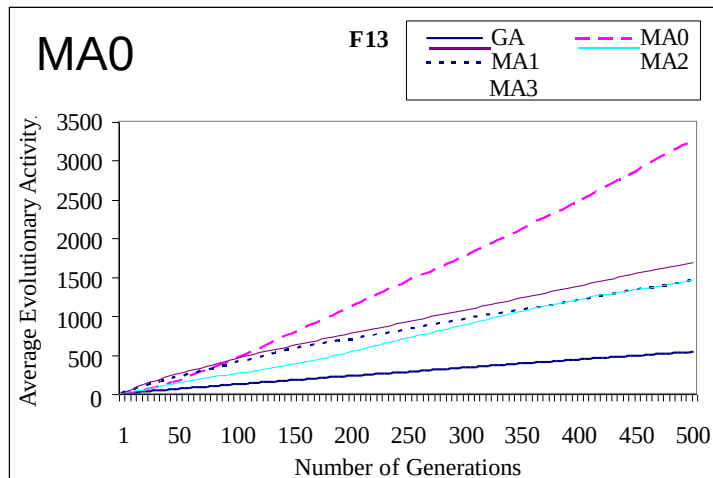
MA3

MA3

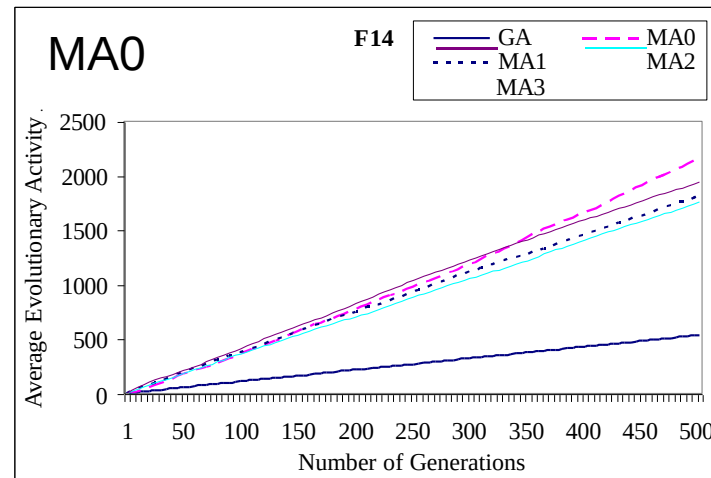


MMA 2.c

MA0



MA0



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Results

- Average evolutionary activity plots show that the multimeme approach successfully identifies useful memes.
 - Even if there are useless operators in the system.
- Any hill climber seems to attain the optimum fast for F1, F3 and F11 under any setting.
- In all runs full success is achieved for all cases.

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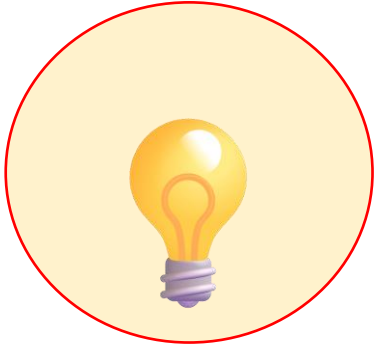


Results

- MMA can identify the best meme or a meme that does not perform significantly better than the best meme for almost each benchmark function.
- Comparing the experimental results obtained using MMA and MA with the best meme for each benchmark function indicate that MA with the best meme is superior based on the average number of evaluations, except for F1, F3, F9 and F11.
 - Learning is time consuming hence there is a trade-off.



Summary



1. Travelling Salesman Problem
2. Generic Permutation based Genetic Operators
3. Multimeme Memetic Algorithms
4. **Benchmark Function Optimisation**



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Questions



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NEXT:

Hyper-heuristics I